

# When is the full spectral reconstruction worth it?

## A showcase landscape for noise-tailored two-qubit gates

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*Following the June 12 discussion: the anchored Tarucha-class model stays as the realistic arm of the paper, and this note builds its counterpart — a noise landscape that is deliberately engineered, feature by measured feature, so that the value of the full spectral reconstruction is not a matter of bookkeeping but visible at a glance. On it, the same blind pipeline as always (the optimizer never sees ground truth; every quoted “true” infidelity is evaluated exactly on the analytic model) delivers entangling gates at the  $10^{-4}$  level on a device whose generic dynamically-corrected gates sit at today’s  $10^{-3}$ – $10^{-2}$ , with noise-tailoring margins that grow from  $2\times$  at the shortest gate to  $44\times$  at  $0.8 T_2^*$  — and a knowledge-ablation ladder showing precisely which ingredient of the characterization pipeline buys what. All numbers June 12, 2026; gate results on the 64k-shot reconstruction unless stated.*

## 1 The landscape: engineered contrast from measured ingredients

The question this scenario is built to answer is the one raised on the thread: *when does knowing the spectrum — all of it — pay dramatically?* The anchored model answered “for that device, modestly” (a flat  $\sim 1.5$ – $1.7\times$ ). The honest way to show the other end of the dial is not to inflate noise strengths but to engineer *contrast*: concentrate the dephasing weight where generic decoupling cannot avoid it, and leave quiet windows that only a spectrally informed design can find. Every ingredient below is a measured *type* of spectral feature; what is stylized is only how starkly they stand apart. The device itself is kept resolutely realistic:  $T_2^* = 3500 \tau$  per qubit ( $17.6 \mu\text{s}$  at the  $\tau = 5 \text{ ns}$  anchor — the measured Ge-hole value, equivalently purified  $^{28}\text{Si}$  at  $\tau \approx 7 \text{ ns}$ ), and a finite control bandwidth, min pulse separation =  $8\tau$  (40 ns  $\pi$ -pulse spacing at the anchor), applied symmetrically to every sequence considered — library, tailored, and counterfactual alike.

The composition (Fig. 1):

- **A quasistatic-class slow bath** on each qubit (correlation time  $\sim 400\tau$ ) carries the entire  $T_2^*$  budget. Its in-band  $\omega^{-2}$  tail is what punishes the *low* decoupling orders: CDD1 and CDD2 cannot escape it ( $\gtrsim 7 \times 10^{-3}$  at  $T_G = 320\tau$ ).
- **A defect-harmonic line family**  $\{\omega_0, 2\omega_0, 3\omega_0, 4\omega_0\}$  with  $\omega_0\tau = 0.051$  on the qubit-local channels. Measured precedent for a defect line plus its harmonic: the  $^{73}\text{Ge}$  Larmor line ( $f_L$ , Hendrickx 2024) and the HRL observation of  $f_L$  and  $2f_L$  surviving isotopic purification. The placement is the engineered part: the de-duplicated CDD ladder at  $T_G = 320\tau$  has fundamentals  $\pi n/T_G$  with  $n = \{5, 10, 21\}$  for CDD3/4/5 — i.e. at  $\{0.049, 0.098, 0.206\}$ , each within a line width of a family member. Doubling the gate time slides the ladder down by exactly one harmonic, so the trap holds across the whole  $T_G$  sweep — that is why the margin *grows* with gate time in Fig. 4 instead of appearing at one sweet spot.
- **A top-window line at  $\omega\tau = 0.372$ .** Under the bandwidth constraint the densest reachable train at  $320\tau$  has its passband at  $\pi \cdot 39/320 = 0.383$ : any design that simply flees up the falling  $1/f^{0.9}$  electrical floor — including every line-blind design — parks exactly there. This line is the price of coarse knowledge.
- **A coupler TLF resonance at  $\omega\tau = 0.236$  on the ZZ channel only** (an independent process on  $\zeta_{12}$ ; Connors-type knee retained there at a low level). Single-qubit QNS cannot see it: it is the structure that only the two-qubit reconstruction reveals, sitting right beside the quiet window the tailored gates use.
- **A quiet electrical floor** ( $1/f^{0.9}$ , the measured charge exponent class) roughly 30 dB below the natural-Si anchored level. This is the one frankly aspirational element — it is the contrast knob — and between the  $4\omega_0$  line’s upper edge and the top line’s lower edge it leaves the *quiet window*  $\omega\tau \in [0.26, 0.31]$  whose floor, not any line, sets what a fully informed design can reach.

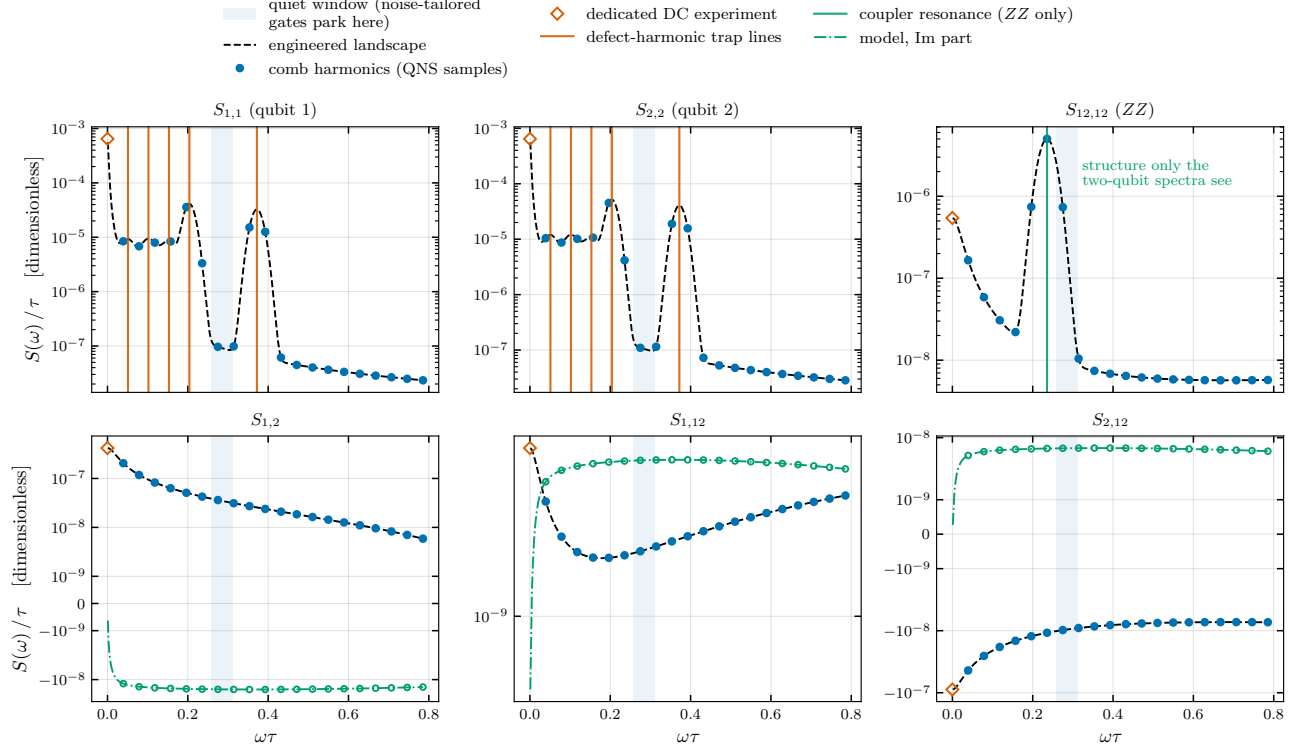


Figure 1: The showcase landscape: the six spectra (dashed), the comb harmonics the QNS protocol samples (dots), the dedicated DC experiments (diamonds), the defect-harmonic trap family (vertical lines), the coupler resonance on  $S_{12,12}$  only, and the quiet window (shaded) in which the noise-tailored gates park. Line centers are off-tooth by construction; the line-aware reconstruction model fits their heights from the teeth they touch.

Two engineering lessons from the probe iterations are worth recording because they are physics, not tuning trivia. First, generic decoupling feels the *area* of a spectral line (its filter lobes are wider than the line), while a tailored design parked nearby feels only the line’s *tails*: narrow-but-tall lines at constant area keep CDD trapped while opening the window — the final landscape uses widths  $\sigma\tau = 0.014\text{--}0.020$ , still resolvable by the comb. Second, there is a structural ceiling: every reachable window must be flanked by mines, and any finite filter lobe integrates a skirt over the total mine area, so the single-gate-time margin saturates ( $\sim 14\times$  at  $320\tau$  here); the way past it is the harmonic lattice, which compounds the punishment with gate time while the tailored gate keeps finding windows —  $44\times$  by  $2560\tau$ .

## 2 Characterization: what 4,000 shots already buy

The protocol is unchanged — the same comb, the same estimators, the same SPAM machinery as the anchored arm — only the landscape underneath moved. Two reconstruction arms were run end-to-end: a 4k-shot arm (the shot-economy question) and a 64k-shot arm (the equal-footing reference for all gate numbers).

**Features.** Every core line tooth (within one line width of a center) is detected at  $> 3\sigma$  with pulls  $\leq 1.9$  in both arms; the coupler line’s teeth on  $S_{12,12}$  are fully resolved at 64k (the 4k arm leaves one shoulder tooth at the consistent-but-not-significant level — the one case the shot escalation was designed for). The double-echo DC experiment lands  $S_{12,12}(0) = 5.48 \times 10^{-7}$  against a true  $5.43 \times 10^{-7}$ .

**The floor is a bound, by design.** In the quiet window the truth sits one to two orders below the shot floor of any realistic acquisition; the teeth there reconstruct as zeros within honest bars (all 63 gap teeth consistent, both arms). This is not a deficiency — it is the operating mode of characterization on a high-contrast landscape: the reconstruction *locates the structure* and *bounds the quiet*, and the bound is what the design needs. The same property makes every prediction conservative (Sec. 3).

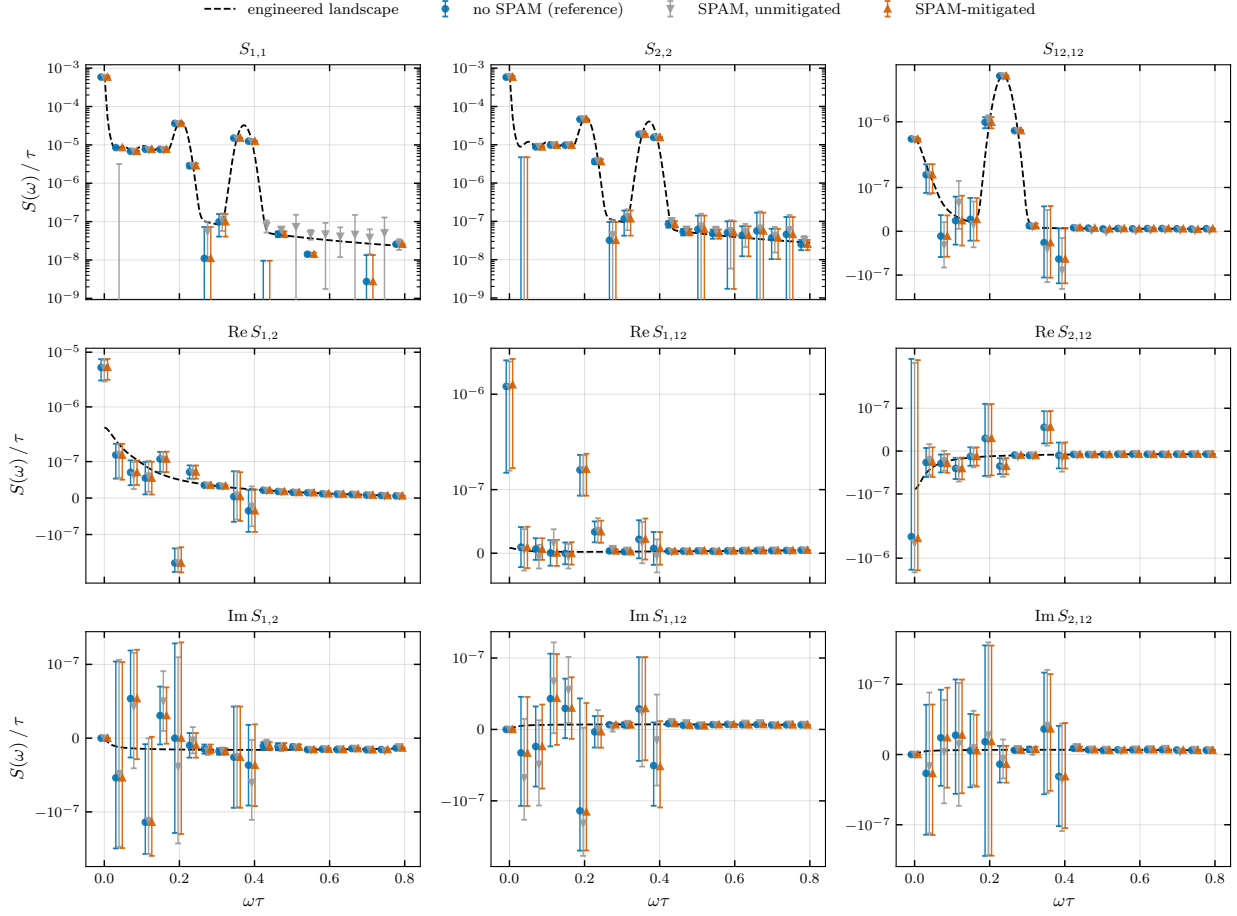


Figure 2: SPAM arms against the engineered truth at 64k shots. Mitigated  $\equiv$  reference everywhere; the raw arm’s bias is resolvable and lands on the worst possible places — the line heights.

**SPAM.** The three SPAM arms (shared phase record, hardware-realistic mild SPAM) behave exactly as in the anchored study: the mitigated reconstruction is indistinguishable from the SPAM-free reference (maximum deviation  $0.03\sigma$  across all six spectra), while the raw arm carries a resolvable bias — up to  $3.8\sigma$  on the  $ZZ$  channel and, notably, a trap-line tooth mis-measured at pull 2.9: on a featured landscape, SPAM bias falsifies *line heights*, i.e. exactly the quantities a tailored design leans on (Fig. 2).

### 3 The knowledge ladder: one device, five designers

The meeting’s central ask was to make the value of the *full* reconstruction visible. The cleanest object we found is a ladder: the same device, the same gate time ( $T_G = 320\tau$ ), the same optimizer and control constraints — only the noise knowledge handed to the designer changes. Every row is the *true* infidelity of the blind winner (Fig. 3a):

| knowledge given to the designer                  | what it stands for      | true $1 - F_{\text{pro}}$ |
|--------------------------------------------------|-------------------------|---------------------------|
| none (best CDD/mqCDD of the library)             | generic DD practice     | $3.8 \times 10^{-3}$      |
| line-blind smooth fit of the same comb data      | coarse characterization | $7.1 \times 10^{-3}$      |
| single-qubit spectra only ( $S_{1,1}, S_{2,2}$ ) | 1Q QNS campaign         | $3.0 \times 10^{-4}$      |
| all six, SPAM-unmitigated (raw arm)              | skipping SPAM treatment | $2.7 \times 10^{-4}$      |
| all six, SPAM-mitigated / reference              | the paper’s pipeline    | $2.7 \times 10^{-4}$      |

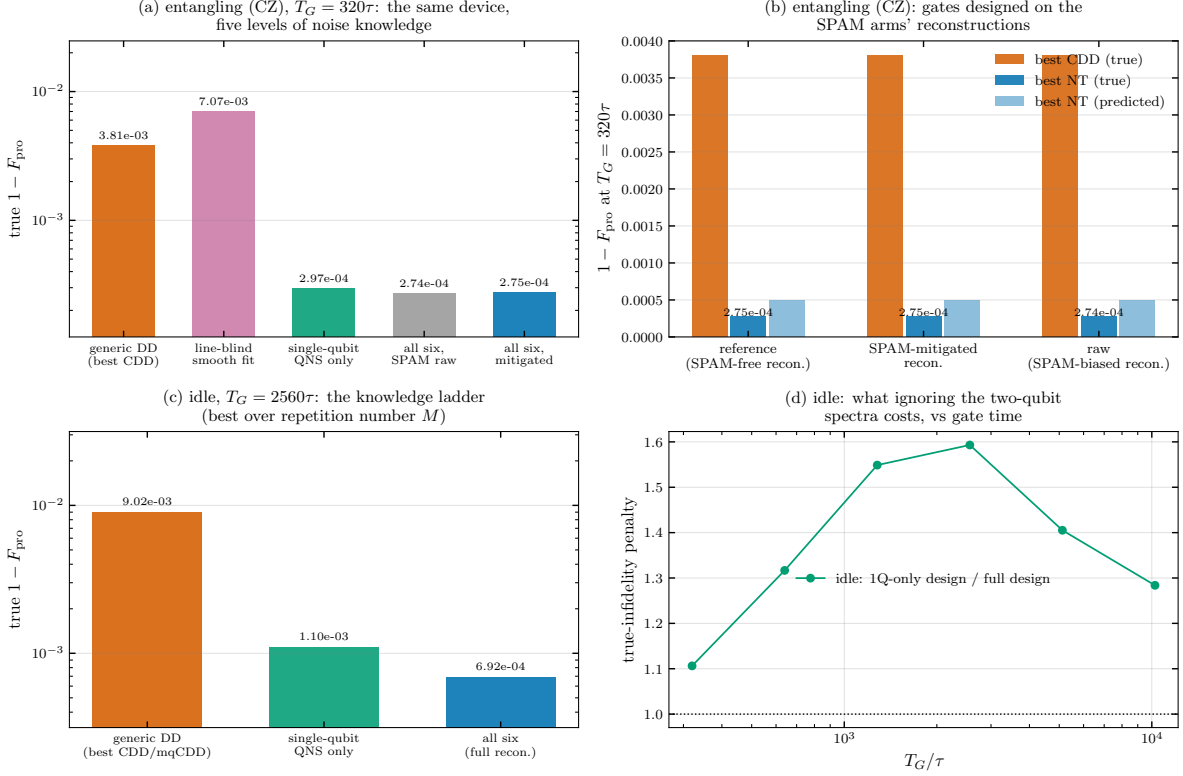


Figure 3: (a) The knowledge ladder at  $T_G = 320\tau$ : true infidelity of the blind winner as the designer’s noise knowledge degrades. (b) Designs on the three SPAM arms: at hardware-realistic mild SPAM the design survives raw bias; the mitigated arm is exactly the reference. (c) The idle-gate ladder at  $T_G = 2560\tau$ . (d) The idle penalty for designing from single-qubit spectra alone, versus gate time.

Three punchlines. (i) *Coarse knowledge is worse than no knowledge*: the smooth-fit designer, armed with the very same measured comb but no line model, confidently parks the filter at the top of its fitted  $1/f$  floor — onto the top-window line — and delivers a gate  $26\times$  worse than the full design and  $1.9\times$  worse than generic CDD. Spectral knowledge is only as good as its resolution of structure. (ii) *The full reconstruction buys  $14\times$  over practice at this gate time* ( $26\times$  over coarse characterization), and the design transfers blind: the same optimizer run on the 4k-shot arm lands the *identical* winner — four thousand shots of characterization fund a  $2.7 \times 10^{-4}$  entangling gate. (iii) *The two-qubit spectra matter exactly where the operation gives the designer ZZ freedom*. For the CZ the single-qubit-only and raw-SPAM designs escape punishment — the gate’s near-diagonal trains happen to put little merged-filter weight on the coupler resonance, and the mild raw bias, though it visibly corrupts a window ranking, is rescued by the optimizer’s refinement; their cost here is *certification*, not design. For the *idle* gate, where synchronizing or staggering the two trains is a genuine design axis, ignoring the two-qubit spectra costs  $1.1\text{--}1.6\times$  in true infidelity across the whole sweep (Fig. 3d, peaking at  $T_G = 2560\tau$ :  $1.10 \times 10^{-3}$  against the full design’s  $6.9 \times 10^{-4}$ ). Finally, all arms over-predict the tailored gate’s infidelity by the same factor 1.8 (the bounded-not-resolved window floor enters as an upper limit), so predictions are uniformly conservative — the pipeline never flatters its own gates.

## 4 The gates: a margin that grows to $44\times$ at realistic infidelities

Figure 4 is the headline. Across the full gate-time sweep, all under the bandwidth constraint and the blind protocol:

| $T_G/\tau$ | FID                  | best CDD             | best NT              | margin       |
|------------|----------------------|----------------------|----------------------|--------------|
| 40         | $1.5 \times 10^{-3}$ | $5.2 \times 10^{-4}$ | $2.5 \times 10^{-4}$ | $2.1\times$  |
| 80         | $5.1 \times 10^{-3}$ | $8.7 \times 10^{-4}$ | $2.8 \times 10^{-4}$ | $3.1\times$  |
| 160        | $1.8 \times 10^{-2}$ | $2.0 \times 10^{-3}$ | $2.9 \times 10^{-4}$ | $6.9\times$  |
| 320        | $6.2 \times 10^{-2}$ | $3.8 \times 10^{-3}$ | $2.7 \times 10^{-4}$ | $13.9\times$ |
| 640        | $1.8 \times 10^{-1}$ | $7.8 \times 10^{-3}$ | $3.3 \times 10^{-4}$ | $23.7\times$ |
| 1280       | $3.9 \times 10^{-1}$ | $1.5 \times 10^{-2}$ | $4.3 \times 10^{-4}$ | $35.4\times$ |
| 2560       | $6.1 \times 10^{-1}$ | $3.1 \times 10^{-2}$ | $6.9 \times 10^{-4}$ | $44.2\times$ |

The two requested properties hold simultaneously. *Realistic infidelities*: the tailored entangling gate sits between  $2.5 \times 10^{-4}$  and  $6.9 \times 10^{-4}$  *everywhere* — on a device whose unprotected coherence would put a  $2560\tau$  gate at 61% error, the tailored gate at the same length is still under  $10^{-3}$ , while the best generic sequence of the library has climbed to  $3 \times 10^{-2}$ . *A dramatic, robust margin*: the CDD-over-NT ratio grows monotonically —  $2\times$ ,  $3\times$ ,  $7\times$ ,  $14\times$ ,  $24\times$ ,  $35\times$ ,  $44\times$  — because the harmonic lattice re-traps the CDD ladder at every doubling while the tailored design keeps finding the windows. Folded through the full reconstruction uncertainty (200-draw ensembles per gate time), the *predicted* margins are conservative and still decisive:  $7.9 \times [7.7, 8.1]$  at  $320\tau$  and  $41 \times [34, 46]$  at  $2560\tau$  (95% CI).

The idle gate tells the same story with repetition number  $M$  as an extra weapon for *both* sides (best over  $M \in \{1, \dots, 128\}$  per gate time, same constraints):

| $T_G/\tau$ | FID                  | best CDD/mqCDD       | best NT              | margin       |
|------------|----------------------|----------------------|----------------------|--------------|
| 320        | $6.2 \times 10^{-2}$ | $1.3 \times 10^{-3}$ | $3.0 \times 10^{-4}$ | $4.4\times$  |
| 640        | $1.8 \times 10^{-1}$ | $2.4 \times 10^{-3}$ | $3.2 \times 10^{-4}$ | $7.6\times$  |
| 1280       | $3.9 \times 10^{-1}$ | $4.6 \times 10^{-3}$ | $4.2 \times 10^{-4}$ | $11.0\times$ |
| 2560       | $6.1 \times 10^{-1}$ | $9.0 \times 10^{-3}$ | $6.9 \times 10^{-4}$ | $13.0\times$ |
| 5120       | $7.3 \times 10^{-1}$ | $1.8 \times 10^{-2}$ | $1.2 \times 10^{-3}$ | $15.2\times$ |
| 10240      | $7.5 \times 10^{-1}$ | $4.5 \times 10^{-2}$ | $1.6 \times 10^{-3}$ | $27.6\times$ |

Tailored idling holds below  $1.7 \times 10^{-3}$  out to  $10240\tau \approx 3T_2^*$ , where generic practice has reached  $4.5 \times 10^{-2}$  —  $28\times$  — and free running has fully dephased. The predicted margins under reconstruction uncertainty are again conservative and decisive:  $3.9 \times [3.6, 4.0]$  at  $320\tau$  rising to  $25 \times [17, 33]$  at  $10240\tau$ .

**Where this leaves the two-scenario story.** The anchored arm says: on a quiet, monotonic, well-behaved device, generic DD is nearly optimal and tailoring buys a reproducible but modest  $1.5$ – $1.7\times$  — do it if a factor approaching two matters to you. The showcase arm says: the moment a device carries resolvable structure — defect harmonics, coupler resonances, the things that purification demonstrably does *not* remove — the same characterization pipeline, at as little as 4k shots per estimator, is the difference between  $10^{-2}$ -class and  $10^{-4}$ -class two-qubit gates, and no intermediate shortcut (coarse fits, single-qubit campaigns, unmitigated SPAM) delivers it. Which device you have is an empirical question — and answering it is itself the reconstruction.

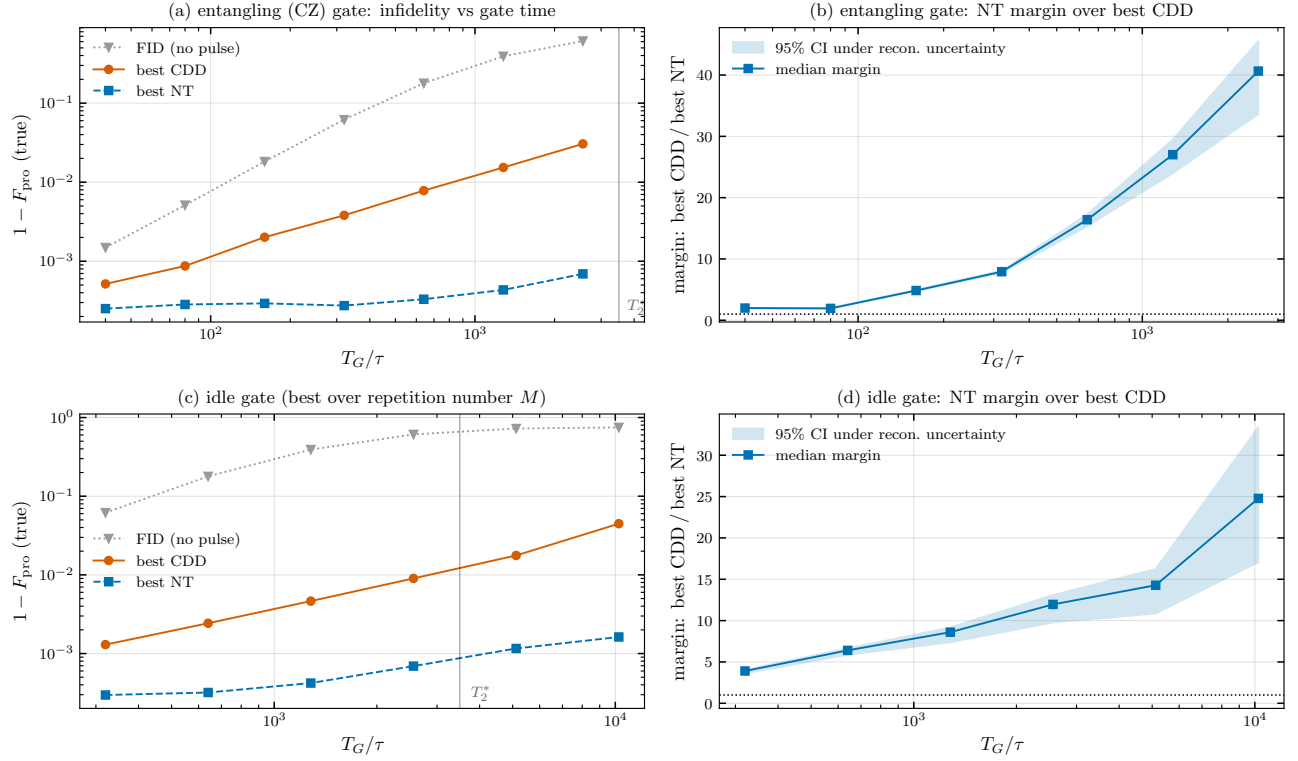


Figure 4: Infidelity versus gate time and the noise-tailoring margin with 95% reconstruction-uncertainty bands, for the entangling (top) and idle (bottom) gates on the showcase landscape, 64k-shot reconstruction, bandwidth-constrained control.